17. If $\dfrac{kx}{(x+2)(x-1)} = \dfrac{2}{x+2} + \dfrac{1}{x-1}$, then the value of k is

(1) 1 (2) 2 (3) 3 (4) 4

18. If $\frac{1-2x}{3+2x-x^2} = \frac{A}{3-x} + \frac{B}{x+1}$, then the value of $A+B$ is

(1) $\frac{-1}{2}$ (2) $\frac{-2}{3}$ (3) $\frac{1}{2}$ (4) $\frac{2}{3}$

19. The number of roots of $(x+3)^4 + (x+5)^4 = 16$ is

(1) 4 (2) 2 (3) 3 (4) 0

20. The value of $\log_3 11 \cdot \log_{11} 13 \cdot \log_{13} 15 \cdot \log_{15} 27 \cdot \log_{27} 81$ is

(1) 1 (2) 2 (3) 3 (4) 4

Summary

- π and $\sqrt{p}$, where p is a prime number, are some irrational numbers.
- $|x-a| = r$ if and only if $r \geq 0$ and $x - a = \pm r$.
- $|x-a| \leq r$ if and only if $-r \leq x - a \leq r$ or $a - r \leq x \leq a + r$.
- $|x-a| > r$ implies $x < a - r$ and $x > a + r$ (or) $x \in (-\infty, a-r) \cup (a+r, \infty)$
- inequalities, in general, have more than one solution.
- The nature of roots of $ax^2 + bx + c = 0$ is determined by the discriminant $D = b^2 - 4ac$.
- A real number a is a zero of a polynomial function $f(x)$ if and only if $(x-a)$ is a factor of $f(x)$.
- If degree of $f(x)$ is less than the degree of $g(x)$, then $\frac{f(x)}{g(x)}$ can be written as sum of its partial fractions.
- In general exponential functions and logarithmic functions are inverse functions to each other.